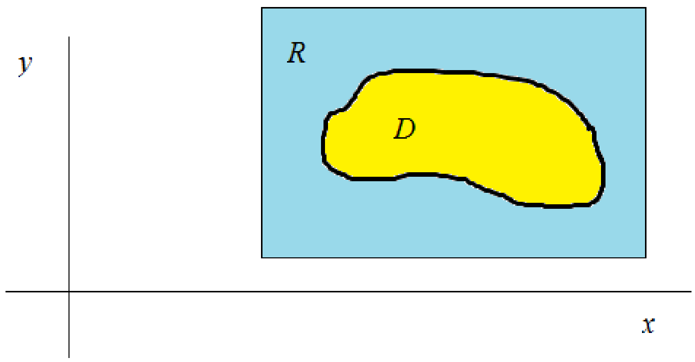
MTH 202 **Chapter 4: Multiple Integrals**

4.2 Double Integrals over Nonrectangular Regions

Motivation: We want to be able to integrate two variable functions not just over rectangles but also over regions *D* of a more general shape.

Suppose we want to integrate the function $f(x, y)$ over a bounded region *D*. Since *D* is bounded we know that it can be enclosed in a rectangle. A picture of what *D* might look like along with a rectangle it is contained in is:

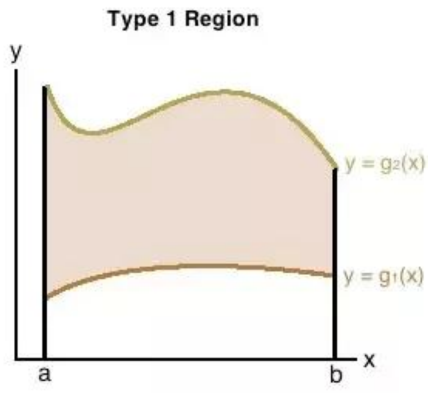


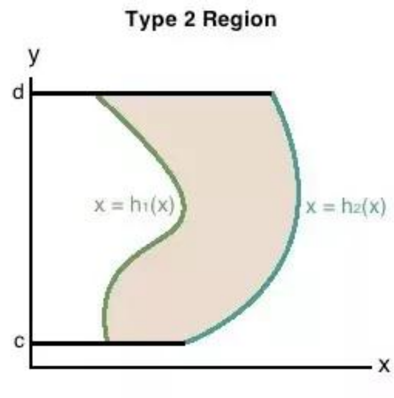
Definition: If F is integrable over R, then we define the <u>double integral of f over D</u> by $\iint_D f(x, y)\,dA = \iint_R F(x, y)\,dA$ (where we are using the notation and symbols introduced in the motivation).

Definition: We say that a plane region *D* is a

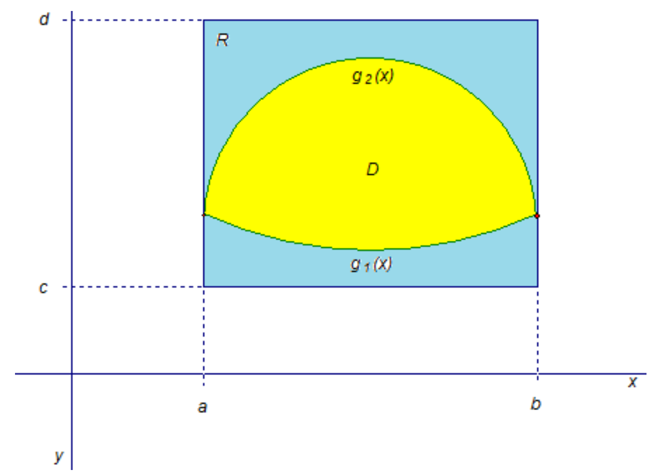
- <u>type 1</u> region if it lies between the graphs of two continuous functions of *x*, that is, $D = \{(x, y) \mid a \le x \le b,\ g_1(x) \le y \le g_2(x)\}$ where $g_1(x)$ and $g_2(x)$ are continuous on $[a, b]$.

- <u>type 2</u> region if it lies between the graphs of two continuous functions of *y*, that is, $D = \{(x, y) \mid c \le y \le d,\ h_1(y) \le x \le h_2(y)\}$ where $h_1(y)$ and $h_2(y)$ are continuous on $[c, d]$.





Exploration:
Suppose we wish to evaluate $\iint_D f(x,y)\,dA$ when $D = \{(x,y): a \le x \le b,\ g_1(x) \le y \le g_2(x)\}$ is a type 1 region and f is continuous on D. To do this, we pick a rectangle $R = [a,b]\times[c,d]$ containing D. A picture of what R might look like is shown below.



We now summarize the results of the exploration above.

Theorem:

1. If f is continuous on a *type 1* region D such that $D=\{(x,y)\,|\,a\le x\le b,\ g_1(x)\le y\le g_2(x)\}$ and both $g_1(x)$ and $g_2(x)$ are continuous on $[a,b]$, then $\iint\limits_D f(x,y)\,dA=\int_a^b\int_{g_1(x)}^{g_2(x)} f(x,y)\,dy\,dx$.
2. If f is continuous on a *type 2* region D such that $D=\{(x,y)\,|\,c\le y\le d,\ h_1(y)\le x\le h_2(y)\}$ and both $h_1(y)$ and $h_2(y)$ are continuous on $[c,d]$, then $\iint\limits_D f(x,y)\,dA=\int_c^d\int_{h_1(y)}^{h_2(y)} f(x,y)\,dx\,dy$.

Question: Could we interchange the order of the integrals and differentials in the most recent theorem?

Example 1: Find the volume of the solid which is in the first octant and lies under the plane given by: $x+3y+2z=6$.

Question: Why would it be incorrect to use the iterated integral $\int_0^2 \left[\int_0^6 \frac{6-3y-x}{2}\, dx \right] dy$ *to find the desired volume in this example?*

<u>Example 2:</u> Suppose that D is the region in the xy-plane bounded by $x = y^2$ and $x = 4$. Use a double integral to find the volume of the solid that lies below the surface given by $z = 1 + x^2 y^2$ and above the region D.

Example 3: Evaluate the iterated integral $\int_0^1 \int_x^1 \sin\left(y^2\right) dydx$.

Example 4: Evaluate $\iint_D xe^{y^2} dA$ over the region D in the first quadrant which is bounded by the graphs $y = x^2$, $x = 0$, and $y = 4$.

Double Integral Properties Over a Region *D*:

Assume that $\iint_D f(x,y)\,dA$ and $\iint_D g(x,y)dA$ exist. Also suppose that c is a constant. Then,

1. $\iint_D \left[f(x,y) + g(x,y) \right] dA = \iint_D f(x,y)\,dA + \iint_D g(x,y)\,dA$
2. $\iint_D cf(x,y)\,dA = c\iint_D f(x,y)\,dA$
3. If $f(x,y) \ge g(x,y)$ for all (x,y) in D, then $\iint_D f(x,y)\,dA \ge \iint_D g(x,y)\,dA$
4. If $D = D_1 \cup D_2$ where D_1 and D_2 don't overlap except perhaps at the boundaries, then $\iint_D f(x,y)\,dA = \iint_{D_1} f(x,y)\,dA + \iint_{D_2} f(x,y)\,dA$.
5. If $A(D)$ represents the area of D, then $\iint_D 1\,dA = A(D)$.
6. If $m \le f(x,y) \le M$ for all $(x,y) \in D$, then $mA(D) \le \iint_D f(x,y)\,dA \le MA(D)$.

<u>Example 5:</u> Suppose that D is the region in the xy-plane bounded by $x = y^2$ and $x = 4$. Use a double integral to find the area of D.